

Esercizio 1. Assume $\mathcal{T}$ irreducible. Then $\mathcal{T}^+$ irreducible.

Proof. Since $\mathcal{T}$ irreducible, let M such that $Th_{pu}(M) = \mathcal{T}$. Then M^+ is ec model of $\mathcal{T}^+$. We claim that $Th_{pu}(M^+) = \mathcal{T}^+$.

By contradiction, let ψ pp sentence such that $M^+ \models \neg\psi$ and $\neg\psi \notin \mathcal{T}^+$. Write ψ as $\exists \vec{x} \bigwedge Q_k(\vec{x})$, which can be decomposed as $\exists \vec{x} \bigwedge_{Q_k=\Phi} \Phi(\vec{x}_k) \wedge \bigwedge_{Q_k \neq \Phi} Q_k(\vec{x}_k)$ where $\vec{x}_k$ denotes the variables effectively occurring in Q_k .

Now denote ψ' the sentence obtained by replacing each occurrence of Φ in ψ with $\exists y\phi$. Since $\neg\psi \notin \mathcal{T}$ we have by definition that $\mathcal{T} \not\models \neg\psi'$. In particular there must be some $N \models \mathcal{T}$ such that $N \models \psi'$ which we can suppose ec using the direct limit construction (ψ' will be preserved since morphism preserve pp formulas).

Now observe that:

- $N^+ \models \psi$ since the semantic of Φ is by definition the same of $\exists y\phi$.
- N^+ is ec model of $\mathcal{T}^+$ since we took N ec model of $\mathcal{T}$.

This is a contradiction since M^+, N^+ are ec models of $\mathcal{T}^+$ but they don't agree on the pp formula ψ . Then letting U be the forgetful functor one expects, $U(M^+), U(N^+)$ are ec models of $\mathcal{T}$ that don't agree on the (equivalent to a) pp formula ψ' . This is a contradiction since $\mathcal{T}$ is irreducible.

□